# One Step Function Explains Why Galaxies Die

*Gas supply versus gas depletion timescales unify black hole and halo quenching across 350,000 galaxies.*

**arXiv:2610.20505 — Step-like quenching from gas mass conservation**

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## Background

Galaxies stop forming stars through many routes, from black hole feedback to halo gas heating. A single framework linking these diverse quenching mechanisms has remained elusive.

**No simple, unified rule connects black hole feedback, halo mass, and galaxy quenching into one predictive framework.**

## What this paper is selling

Deriving a step-function quenching response from gas mass conservation, the authors infer gas-supply timescales directly from observed quenched fractions. This yields a characteristic 1 Gyr timescale and separates black hole from halo regulation.

## Key results to remember

- Quenched fractions imply a characteristic gas-supply timescale of about 1 Gyr.
- Black holes drive inside-out quenching; dark matter halos drive outside-in quenching.
- Intermediate surface density galaxies sit at a critical quenching and rejuvenation state.

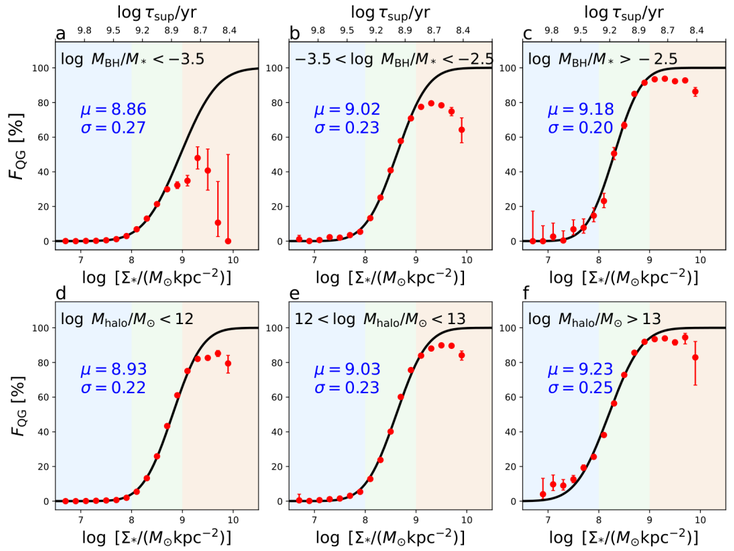



**Figure 2 |** Black hole growth lengthens and narrows supply timescales, while halo mass lengthens and broadens them, separating their quenching roles.

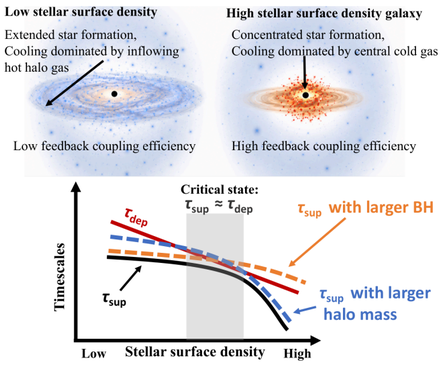



**Figure 4 |** At intermediate surface density, supply and depletion timescales converge, making galaxies most sensitive to quenching and rejuvenation.